

Mahmoud Youssef

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TECHNICAL SKILLS

Languages: Java, SQL

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, MapStruct, JdbcTemplate

Databases: MySQL, PostgreSQL, Redis (caching, rate limiting, session/TTL state)

Security: JWT Authentication, Refresh Token Rotation, RBAC, HMAC-SHA256 Signing

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Maven, Flyway, Swagger/OpenAPI, Postman, JUnit 5, Mockito

Methodologies: Object-Oriented Programming (OOP), SOLID, Clean Architecture, RESTful API Design, Agile/Scrum

PROJECTS

CalCounter — AI-Powered Nutrition Tracking | *Spring Boot 3, MySQL, JWT, HuggingFace, Flutter, Docker*
GitHub | Live App | API Docs

- Graduation project; led backend in a cross-functional team (backend, Flutter, AI); built 30+ endpoint REST API with **JWT/RBAC** via **Spring Security**, deployed on Render (**Docker + MySQL**) with Flutter web on Firebase Hosting
- Reduced food photo analysis wait time ~60% by integrating HuggingFace EfficientNetB3 Food-101 model via async pipeline (**Spring @Async**); frontend polls status endpoint for result, replacing blocking synchronous flow

Attendance Management API — QR-Based Workforce Tracking | *Spring Boot 4, MySQL, Redis, JWT, Docker* GitHub

- Designed HMAC-SHA256 signed, time-boxed QR check-in sessions; per-employee dedup via **Redis** keys (not a global one-time flag), allowing one signed session to serve an entire team while blocking duplicate check-ins per employee
- Built attendance engine with schedule-driven status calculation (PRESENT/LATE/HALF_DAY) and a daily **@Scheduled** job auto-marking unrecorded employees ABSENT; leave approval auto-reconciles **attendance_records** for the full date range
- 34-endpoint REST API across 8 controllers with method-level **RBAC** (**@PreAuthorize**); reporting layer uses native SQL via **EntityManager**, including a recursive CTE for date-range absence reports with zero N+1 queries

LogiBridge — B2B Order Management Platform | *Spring Boot 3, PostgreSQL, Redis, Docker* GitHub

- Built 15+ endpoint order-management API; multi-role **RBAC** via **Spring Security @PreAuthorize**; **JWT** + refresh token rotation with reuse detection; immutable audit trail via **@PreUpdate** hook
- Exactly-once state transitions via **PostgreSQL** unique constraint + pessimistic locking with optimistic **@Version** fallback; idempotency keys (DB-backed) for safe network retries; validated with **JUnit 5** + Mockito

E-Commerce Backend System | *Spring Boot 3, MySQL, Redis, Stripe, Docker* GitHub

- 30+ endpoint REST API; decoupled order/payment flows via domain events (**OrderCreatedEvent**, **PaymentCompletedEvent**); applied SOLID + Clean Architecture across all modules
- Idempotent Stripe webhook (HMAC); two-phase inventory reservation + optimistic locking; **Redis** cache-aside cut read latency ~40%; SQL double-entry bookkeeping for revenue reporting

University Timetable Scheduler | *Spring Boot 3, MySQL, React, TypeScript, Docker* GitHub

- Full-stack scheduling system; backend enforces 7 hard + 5 soft constraints with async generation and status polling; **JWT** + role-based access (ADMIN / SCHEDULER / INSTRUCTOR)
- React/TypeScript SPA with interactive weekly grid, multi-format export (PDF/Excel/PNG); CI/CD via GitHub Actions; **Docker** Compose deployment

Distributed URL Shortener | *Spring Boot 3, MySQL, Raw JDBC, Docker* GitHub

- **Redis** INCR + Base62 distributed ID generation with zero collisions; **AtomicLong** fallback on Redis failure; atomic per-IP rate limiting via **Redis**; Raw JDBC for explicit query control

EDUCATION

Bachelor of Science (B.Sc.), Computer Science

Expected Jun 2026

El Shorouk Academy, Cairo, Egypt

Relevant Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Systems, Operating Systems, SDLC

CS50x — Introduction to Computer Science

2024

Harvard University (via edX / ITI)